

COMPUTED TOMOGRAPHIC FEATURES OF SKULL OSTEOMYELITIS IN FOUR YOUNG DOGS

GABRIELA SEILER, FEDERICA ROSSI, MASSIMO VIGNOLI, RACHEL CIANCIOLO, TIFFANY SCANLON, URS GIGER

Computed tomography (CT) features of four immature to young adult dogs with osteomyelitis of the skull are described. Trauma or bite wounds were the cause of infection and *Staphylococcus aureus* was the most common pathogen. CT features were a combination of soft tissue thickening, bone lysis, and bone proliferation. Bone lysis was extensive in some dogs with a moth-eaten appearance and involved the calvarium, base of the skull, the frontal sinuses, and the temporomandibular joint. In other dogs it was more focal with thinning of the bone rather than complete lysis. Bone proliferation also varied in appearance from irregular palisading or spiculated to expansion and septation of the frontal bone. Sequestrum formation was seen in one dog. Widespread infection in one dog involved the tympanic bullae and the temporomandibular joint. Lysis of the calvarium resulted in bacterial meningitis in two dogs. One dog was euthanized and three were treated with surgical curettage of the affected bone and antibiotic therapy which resulted in resolution of the clinical signs in one dog whereas two dogs had recurrent disease. CT was very helpful for characterizing extent and localization of the infection. Despite the aggressive CT features, osteomyelitis should be considered especially in young animals with a history of trauma or bite wounds. The pathophysiology of skull bone infections is discussed. *Veterinary Radiology & Ultrasound*, Vol. 48, No. 6, 2007, pp 544–549.

Key words: computed tomography, dog, meningitis, osteitis, osteomyelitis, skull.

Introduction

OSTEOMYELITIS MAY ORIGINATE primarily in bone or result from secondary seeding. The term osteomyelitis implies the presence of bone marrow infection but it is widely used to describe bacterial infections of flat bones of the skull and for the purpose of this paper will be used for all bones of the skull. The most common sites of osteomyelitis are the long bones and vertebrae.¹ In hematogenous osteomyelitis in growing animals, bacteria migrate to the metaphyses and epiphyses of long bones where sluggish and turbulent blood flow in the venous sinusoids in the growing bone facilitates bacterial seeding.^{2,3} Common routes of infection in adult animals are deep injuries caused by penetrating foreign bodies, trauma, bite wounds, and surgical interventions.^{3–6} Open fractures and osteosynthesis with surgical implants are particularly frequent causes for osteomyelitis in long bones.⁷ However, osteomyelitis may also be secondary to soft tissue infections, such as endocarditis, pyometra, and prostatitis, with subsequent seeding into bone via blood and lymph circulation.

The radiographic appearance of osteomyelitis in the appendicular and axial skeleton in dogs is understood.^{1,8,9} However, descriptions of skull osteomyelitis are infrequently reported in dogs, and no reference has been made to the computed tomography (CT) findings.^{10–12} Osteomyelitis of the skull is difficult to assess radiographically due to the complex anatomy and superimposition of structures. While the value of CT for assessment of nasal and sinusoidal infections, and skull tumors is understood,^{13–15} we describe here the CT features of osteomyelitis of the skull in four immature to young adult dogs.

Materials and Methods

CT images of four dogs with diagnosis of a bacterial infection of the skull based on culture from either biopsy or necropsy were retrospectively analyzed. CT scans were performed using a single-slice helical CT unit* either at the Matthew J Ryan Veterinary Hospital at the University of Pennsylvania (VHUP) or Clinica Veterinaria dell'Orologio (CVO). Contiguous slices of the entire head were acquired with a detail algorithm using 3 mm (VHUP) or 5 mm (CVO) slice thickness and with a bone algorithm using 1 mm (VHUP) or 5 mm (CVO) slice thickness. Nonionic intravenous contrast medium† was used in one dog (dog #2) at a dosage of 800 mg iodine/kg.

From the Department of Clinical Studies-Philadelphia, School of Veterinary Medicine, University of Pennsylvania, 3900 Spruce Street, Philadelphia, PA 19104-6010 (Seiler, Cianciolo, Scanlon, Giger) and Clinica Veterinaria dell'Orologio, Via Gramsci, 40037 Sasso Marconi, Bologna, Italy (Rossi, Vignoli)

Address correspondence and reprint requests to Gabriela Seiler, at the above address. E-mail: gseiler@vet.upenn.edu

Received January 12, 2007; accepted for publication May 10, 2007.
doi: 10.1111/j.1740-8261.2007.00293.x

*GE ProSpeed®, General Electric Company, Milwaukee, WI.
†Iohexol 350, Amersham Health Inc., Oslo, Norway.

Results

All four dogs were young to young adult with an age range of 5 months to 3 years, three were intact males and one intact female (Table 1). Three dogs had a history of a dog bite wound (only suspected in dog #1) and one had previous trauma, hitting the head against a hard surface while playing (dog #3). The main presenting complaints were progressive or intermittent swelling, pain, fever, seizures (dog #2), and a fistulous tract (dog #4). Furthermore dog #1 had a 2-month history of failure-to-thrive, progressive inability to open his mouth, severe lethargy, weight loss, and weakness (Table 1).

In dog #1, radiographs of the head were acquired and there was soft tissue thickening of the left side of the head and massive lytic and proliferative changes involving the left temporomandibular joint, the left temporal, and parietal bone. Osteomyelitis and neoplasia were the main initial differential diagnoses. Radiographs were not acquired in other dogs.

CT findings in all dogs were a combination of soft tissue thickening, bone lysis, and bone proliferation to various degrees. Increased soft tissue attenuating material was seen in dog #1 in the tympanic bullae and external ear canals. In dog #1, the frontal sinus was partially filled with soft tissue attenuating material presumed to be fluid based on the meniscus-like appearance of the soft tissue–air interface (Fig. 1A). Soft tissue attenuating material was also seen in the frontal sinuses in dog #2.

Bone lysis was extensive in some dogs such as dog #1 where the calvarium and base of the skull had large areas of bone lysis, some with a moth-eaten appearance, and there were large bone defects present in the left temporal and parietal bone and the ventral wall of the left frontal sinus. In the left temporomandibular joint, the mandibular fossa of the zygomatic arch was particularly eroded, and the joint space irregularly widened (Fig. 1B).

In dog #2 the cribriform plate was also eroded. Bone destruction was more focal in dogs #3 and #4, especially in dog #4 the bone was thinned rather than completely destroyed (Fig. 2).

Bone proliferation had a different appearance in all dogs. It was extensive in dog #1 with a palisading and spiculated appearance (Fig. 1), whereas in dog #2 the predominant feature was enlargement and increased septation of the frontal sinuses (Fig. 3). Bone thickening with a

slightly irregular surface was seen in dog #3 (Fig. 4). The bone proliferation in dog #4, which had the shortest duration of disease, was mild (Fig. 2).

A sequestrum was seen in dog #3, where an irregular, isolated bone fragment surrounded by hypoattenuating material was present in the left caudal frontal bone (Fig. 4A).

Extension of the lesion into the calvarium was present in two dogs (#1 and #2). In dog #1 the soft tissue thickening of the left lateral aspect of the head extended through large areas of bone lysis of the left calvarium compressing the brain as evidenced by a midline shift and compression of the left lateral ventricle (Fig. 1D). In dog #2 there was a large defect present in the cribriform plate (Fig. 3A). After intravenous contrast medium administration there was peripheral uptake, moderate irregular meningeal contrast enhancement, and extension of contrast enhancing tissue into the calvarium, compressing the right frontal lobe and causing a midline shift to the left (Fig. 3B).

In two dogs with recurrent disease, follow-up CT imaging was performed. In dog #2 with a suspected relapse after 1 year, there was an enlarged right frontal sinus, with a multilobulated appearance, filled with nonenhancing soft tissue attenuating material consistent with fluid (Fig. 5). The frontal bone was irregular with new bone formation and small areas of osteolysis. There was also some involvement of the left frontal sinus. Dog #3 had recurrence 3 months after initial treatment. In CT images there was ossification of the left frontal bone, residual thickening, and periosteal reaction. A hypoattenuating line was now visible through the frontal/parietal bone in a dorsolateral to ventromedial direction, likely representing a fracture (Fig. 4B). In the middle of the ossified frontal bone, two small hypoattenuating areas were seen surrounding osseous fragments, consistent with two small sequestra.

The owners of dog #1 elected to euthanize the dog due to the perceived severe pain, a likely poor prognosis as well as financial constraints, and permission for postmortem examination was granted. At necropsy, the left side of the calvarium was irregularly thickened (8 mm) compared with the right side (4 mm). The left temporal bone, parietal bone and basisphenoid bones were multifocally soft, friable, and covered in a viscous, purulent exudate. There were foci of hemorrhage within the left temporalis muscle. All adjacent lymph nodes were enlarged, edematous, and hemorrhagic. The brain was also edematous, and there was compression

TABLE 1. Signalment, History, Duration of Disease Before Initial Presentation, Culture Results, and Outcome of Dogs with Osteomyelitis of the Skull

Case No.	Breed	Sex	Age (Months)	History	Disease Duration	Culture	Outcome
1	Fila Brasileiro	M	7	Bite wound?	2 months	<i>Staphylococcus aureus</i>	Euthanasia
2	German Shepherd	M	5	Bite wound	N/A	<i>S. aureus</i>	Recurrence after 1 year
3	Mixbreed	F	7	Trauma	2 months	<i>S. aureus</i> and <i>epidermidis</i>	Recurrence after 3 months
4	Segugio Italiano	M	36	Bite wound	10 days	Cytology: cocci	Resolved after initial treatment

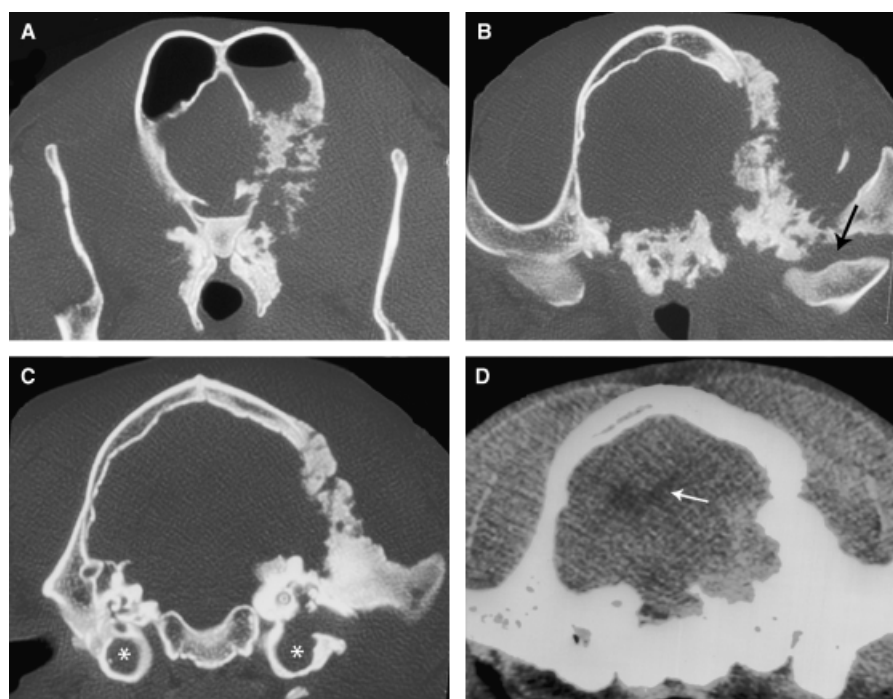


FIG. 1. Transverse computed tomography images of the skull of a 7-month-old male Fila Brasileiro (dog #1) in a bone window (A–C) and soft tissue window (D). There is extensive multifocal lysis of the left frontal, temporal, and parietal bone as well as the left zygomatic arch. The calvarium and base of the skull are thickened and have irregular spiculated periosteal reactions. The left frontal sinus is partially filled with fluid (A). The left temporomandibular joint space (black arrow) is widened and the joint surface is eroded (B). Both tympanic bullae are thickened and filled with soft tissue attenuating material (*) (C). In the soft tissue window (D) a mass effect on the left cerebrum is evident as shift of the midline to the right and compression of the left lateral ventricle (white arrow).

of the left side of the cerebrum. The meningeal space contained a gray–white, viscous exudate.

Histopathologically, there were multifocal to coalescing aggregates of epithelioid macrophages, multinucleated giant cells, lymphocytes, and eosinophils that surrounded large plaque-like colonies of coccoid bacteria. The colonies

of bacteria were circumferentially surrounded by amorphous eosinophilic material, a manifestation of the Splendore–Hoepli phenomenon.¹⁶ The inflammatory cells were primarily neutrophils with fewer macrophages, lymphocytes, and plasma cells. The surrounding bone was often necrotic or remodeled with numerous foci of woven bone proliferation. Extensive fibrosis of both the bone and medullary sinuses was present. There was also evidence of necrosis and regeneration of adjacent musculature. In sections of brain there was severe chronic suppurative meningitis with rare embedded coccoid bacterial colonies (Gram positive); *Staphylococcus aureus* was cultured from the meninges and the tympanic bullae.

Dogs #2–4 were treated with surgical curettage of the affected bone and 4 weeks of antibiotic treatment (cephalexine 20 mg/kg PO BID in dogs #2 and #3, enrofloxacin 5 mg/kg PO SID in dog #3). In dog #4 no signs of recurrence were detected at physical examination after 3 months. In dog #2 a relapse occurred after 1 year and repeated surgical curettage followed by antibiotic therapy resulted in complete resolution of clinical signs after 3 months. Dog #3 had recurrent disease after 3 months, received antibiotics again, but was lost for further follow-up. In two dogs *S. aureus* (dog #2) and *S. aureus* and *epidermidis* (dog #3) was cultured from the removed material. In dog #4 inflammatory cells with intracellular cocci were found, but a culture was not done.



FIG. 2. Transverse computed tomography image of a 3-year-old male Segugio italiano (dog #4) with a history of a previous bite wound and a swelling on the dorsal nasal area. There is diffuse soft tissue swelling on the dorsal and right aspect of the nose. The left nasal bone is interrupted, several bony fragments are present. The borders of the bone have irregular periosteal reaction. A small area of soft tissue attenuating material is present in the dorsal nasal meatus. There is also irregular thinning of the right maxilla (arrows).

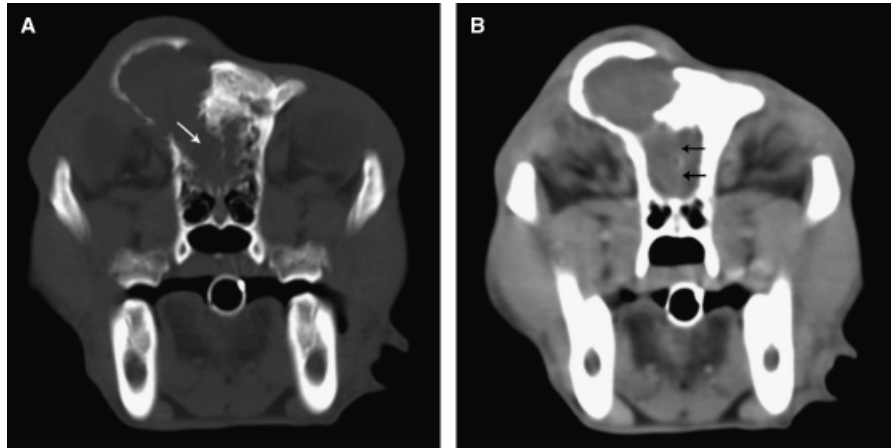


FIG. 3. Transverse computed tomography image of a 5-month-old male German shepherd dog (dog #2) in a bone window (A) and soft tissue window after intravenous contrast medium administration (B). There is soft tissue thickening over the right frontal sinus. The right frontal bone is thinned, expanded, and interrupted in several places. The right dorsal aspect of the cribriform plate is lytic as well (white arrow). The frontal sinus is filled with soft tissue-attenuating material. The frontal bone delineating the left frontal sinus is thickened and irregular, the air space of the sinus is obliterated. There is contrast enhancement of the material within the right frontal sinus as well as the meninges covering the right frontal lobe, and a mild midline shift to the left can be seen (black arrows).

Discussion

The radiographic appearance of osteomyelitis consists of multifocal bone lysis with poorly defined cortical margins, sclerosis, irregular periosteal reactions and formation of sequestra,^{8,17,18} but interpretation of skull radiographs is often difficult. CT is superior to conventional radiographs as it can demonstrate abnormalities earlier and in more detail and does not have the disadvantage of superimposition of the complex anatomic structures of the skull. CT has become the method of choice to assess osteomyelitis of the skull in people.^{17,19}

In two dogs, the calvarium was destroyed resulting in spread of the infection to the meninges which was confirmed in dog #1, whereas in dog #2 meningitis was suspected based on meningeal enhancement in CT images.

Intravenous contrast medium should be administered routinely in dogs where the calvarium is affected, to assess spread of infection to the meninges or brain.

CT can allow detection of the presence of even small sequestra, as in dog #3, which is important as this influences the treatment and outcome.^{18,20} Osseous fragments without blood supply cannot be degraded by inflammatory cells and often resist antibiotic therapy, as in dog #3. In the periphery of the lesion there is reactive hyperemia and increased osteoclastic activity leading to progressive bone lysis, visible as hypoattenuating rim around the bony fragment on CT images. Eventually sequestra become surrounded by fibrous and granulation tissue and finally periosteal new bone. Surgery may be needed to correct these sequestra.

Periosteal proliferations and thickening of the calvarium were present to variable degrees in all dogs in this series,

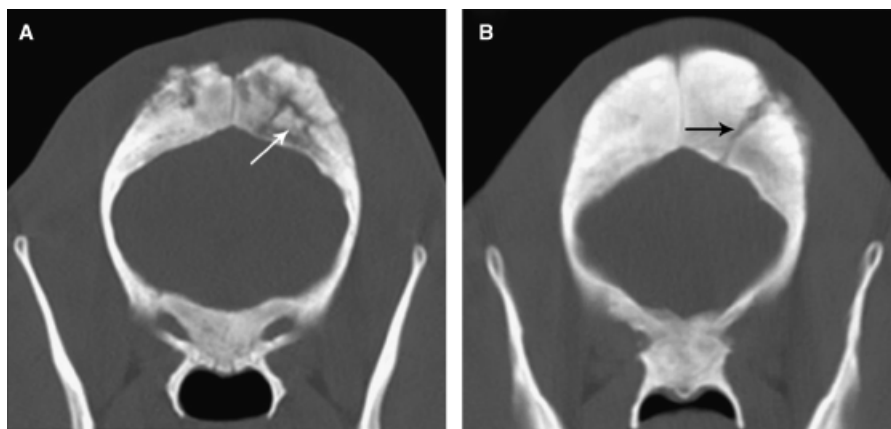


FIG. 4. (A) Transverse computed tomography (CT) image of a 7-month-old female mixed-breed dog (dog #3) presented 2 months after head trauma. Bilateral irregular thickening of the frontal bone (left > right) with periosteal proliferation and small osteolytic areas is seen, there is soft tissue swelling over the left frontal bone present as well. On the left side there is an irregular bony fragment seen within the thickened frontal bone, surrounded by hypoattenuating material (white arrow). A sequestrum was surgically removed. (B) Follow-up CT 3 months after initial presentation. Both frontal bones are severely thickened; a fracture line is now visible in the left frontal bone (black arrow). There is irregular periosteal reaction on the surface of the left frontal bone.



FIG. 5. Transverse computed tomography image of the same dog as in Fig. 3 (dog #2), 1 year after initial presentation. The right frontal sinus is enlarged with a multilobulated appearance and filled with nonenhancing soft tissue-attenuating material (fluid). The frontal bone is irregular with new bone formation and small areas of osteolysis. The left frontal sinus is obliterated and the frontal bone thickened and irregular.

and were likely associated with chronicity as they were more pronounced in dogs with a prolonged history. The most pronounced thickening of the calvarium was present in dog #3 with chronic osteomyelitis of 2-month duration. Callus formation around the skull fracture may have contributed, and it remains unclear if the radiolucent fracture line in dog #3 at the 3-month follow-up exam represents a new fracture or was not found originally due to osteomyelitis and sequestrum formation or CT technique (slice thickness) used. In people, sclerosis and changes in bone width, either thinning or thickening are seen more commonly with more chronic disease as well as in patients with disease resistant to treatment.²⁰

In the dogs described in this series, the CT features of osteomyelitis of the skull were very aggressive with extensive bone lysis, osseous proliferations, and associated soft tissue thickening. Because all dogs were young (5 months to 3 years) and most had a history of trauma or bite wounds, infection was the most likely differential diagnosis. In dog #4, the presence of bone infection is less well documented than in the other dogs due to lack of bacterial culture evaluation. Differential diagnoses for the changes in the CT images in this dog include comminuted fracture with hemorrhage and diffuse soft tissue neoplasia eroding the bone. However, based on the history of a bite wound, presence of bacteria in the surgically removed bony material and the response to therapy, osteomyelitis is still the most likely differential diagnosis.

While very uncommon in young dogs, neoplasia may need to be considered as a differential diagnosis. Bone tumors described in young dogs are lymphoma²¹ and osteosarcoma.²² Furthermore, Ewing's sarcoma, a highly malignant primitive neuroectodermal tumor of chil-

dren^{23,24} has been reported in a puppy.²⁵ CT findings of neoplastic processes of the skull such as squamous cell carcinoma, osteosarcoma, and fibrosarcoma^{26,27} can be difficult to differentiate from osteomyelitis. Permeative bone destruction, cortical bone expansion, and enlargement of the adjacent musculature are reported most consistently with malignant neoplasia in humans, whereas osteomyelitis was more often characterized by sclerosis and periosteal reactions.²⁸ However, there is considerable overlap in imaging findings, as both osteomyelitis and neoplasia can cause sclerosis and/or lysis, and therefore biopsy of lesions for histopathologic examination and aerobic and anaerobic bacterial and fungal cultures are generally a necessity. While in the patients presented here surgical biopsies were obtained, CT guidance could also be used to obtain biopsy samples of specific areas of the lesion.

Trauma and bite wounds were the most common causes in this series of skull osteomyelitis. Bite wounds, as the most likely cause of the skull infection in three dogs reported here, are prone to cause infections. Bacteria are directly introduced into the subcutaneous and deeper soft tissue, and the associated soft tissue damage and devitalization due to crushing and shearing forces provide a favorable environment for microorganisms to colonize. Presence of a soft tissue infection next to a bone, however, does not necessarily lead to osteomyelitis as bone is relatively resistant to infection. Vascular compromise, caused by obstruction of blood vessels from external trauma and accumulations of exudate, likely facilitates bone infection,²⁹ and ensuing thrombi, hematomas, and necrotic tissue are ideal media for bacterial growth.³⁰ Small foreign particles may also pose a risk for persistent and recurrent infections as they evade the host immune response.⁹ Another potential source of skull osteomyelitis is otitis media, when the infection extends into the surrounding soft tissues and eventually invades the skull bone. In dog #1 it is unclear if otitis media was the origin of skull osteomyelitis or if it was a sequel of widespread infection of the head. In people, osteomyelitis of the skull base is rare but can be a serious condition, usually seen with pseudomonas infection in advanced stages of otitis externa.^{31–33} Intracranial extension of otitis media leading to brain stem abscesses has been described in cats.³⁴

S. aureus infection was either documented or suspected in the dogs reported here, and is the most common pathogen causing osteomyelitis of the skull and other sites in humans. The propensity of *S. aureus* to cause osteomyelitis is related to their binding to surface receptors for bone matrix, which aids bacterial adherence.³⁵ Spread of infection is promoted by the production of staphylococcal toxins which destroy host cells, and of their hydrolases which degrade components of the extracellular matrix. *S. aureus* can also promote its uptake by epithelial and endothelial cells and survive in the intracellular environment.³⁶ Both *S. aureus* and *epidermidis* can produce a mucoid polysaccha-

ride biofilm, which reduces the access of antimicrobial agents by providing a diffusion barrier and bacterial phagocytosis.³⁷ Histopathologically, Splendore–Hoeppli phenomenon was observed in dog #1. This phenomenon is most often associated with botryomycosis caused by *S. aureus* and *Actinomyces* sp., but also occurs with other microbial infections and foreign body reactions.¹⁶

Conclusion

CT findings of skull osteomyelitis are characterized by substantial osteolysis, new bone formation and soft tissue

swelling in dogs, similar to that in human patients. Invasion of the calvarium with meningitis, one of the complications associated with skull osteomyelitis, can also be seen. CT was instrumental in characterizing extent and localization of the infection in the dogs of this series. *S. aureus* seems to be a common pathogen causing skull osteomyelitis in dogs just as in humans. Despite the very aggressive CT features, osteomyelitis rather than bone tumors should be considered especially in young animals with a history of trauma or bite wounds but tissue biopsy and bacterial culture are still needed to confirm the diagnosis.

REFERENCES

1. Thomas WB. Diskospondylitis and other vertebral infections. *Vet Clin North Am Small Anim Pract* 2000;30:169–182.
2. Caywood DD, Wallace LJ, Braden TD. Osteomyelitis in the dog: a review of 67 cases. *J Am Vet Med Assoc* 1978;172:943–946.
3. Bubenik LJ. Infections of the skeletal system. *Vet Clin North Am Small Anim Pract* 2005;35:1093–1109.
4. Mandal S, Berendt AR, Peacock SJ. *Staphylococcus aureus* bone and joint infection. *J Infect* 2002;44:143–151.
5. Ameh EA, Shebu BB. Acute (hematogenous) osteomyelitis of the skull in childhood. *West Afr J Med* 2001;20:78–80.
6. Swinebroad EL, Dabareiner RM, Swor TM, et al. Osteomyelitis secondary to trauma involving the proximal end of the radius in horses: five cases (1987–2001). *J Am Vet Med Assoc* 2003;223:486–491.
7. Fossum TW, Hulse DA. Osteomyelitis. *Semin Vet Med Surg (Small Anim)* 1992;7:85–97.
8. Wrigley RH. Malignant versus nonmalignant bone disease. *Vet Clin North Am Small Anim Pract* 2000;30:315–347.
9. Johnson KA. Osteomyelitis in dogs and cats. *J Am Vet Med Assoc* 1994;204:1882–1887.
10. Lyon KF, Bard RA. Mandibular osteomyelitis caused by *Coccidioides immitis*. *J Vet Dent* 1988;5:5–6.
11. Muir P, Johnson KA. Anaerobic bacteria isolated from osteomyelitis in dogs and cats. *Vet Surg* 1992;21:463–466.
12. Grahn BH, Szentimrey D, Battison A, Hertling R. Exophthalmos associated with frontal sinus osteomyelitis in a puppy. *J Am Anim Hosp Assoc* 1995;31:397–401.
13. Lefebvre J, Kuehn NF, Wortinger A. Computed tomography as an aid in the diagnosis of chronic nasal disease in dogs. *J Small Anim Pract* 2005;46:280–285.
14. Saunders JH, van Bree H. Comparison of radiography and computed tomography for the diagnosis of canine nasal aspergillosis. *Vet Radiol Ultrasound* 2003;44:414–419.
15. De Rycke LM, Saunders JH, Gielen IM, van Bree HJ, Simoens PJ. Magnetic resonance imaging, computed tomography and cross-sectional views of the anatomy of normal nasal cavities and paranasal sinuses in mesaticephalic dogs. *Am J Vet Res* 2003;64:1093–1098.
16. Gross TL, Ihrke PJ, Walder EJ. Infectious nodular and diffuse granulomatous and pyogranulomatous diseases of the dermis. In: Reinhardt RW (ed): *Veterinary dermatopathology: a macroscopic and microscopic evaluation of canine and feline skin disease*. St. Louis: Mosby Inc., 1992; 163–169.
17. Sammak B, Abd El Bagi M, Al Shaheed M, et al. Osteomyelitis: a review of currently used imaging techniques. *Eur Radiol* 1999;9:894–900.
18. Schulze D, Blessmann M, Pohlenz P, Wagner KW, Heiland M. Diagnostic criteria for the detection of mandibular osteomyelitis using cone-beam computed tomography. *Dentomaxillofacial Radiol* 2006;35:232–235.
19. Dudkiewicz M, Livni G, Kornreich L, Nageris B, Ulanovski D, Raveh E. Acute mastoiditis and osteomyelitis of the temporal bone. *Int J Pediatr Otorhinolaryng* 2005;69:1399–1405.
20. Ida M, Watanabe H, Tetsumura A, Kurabayashi T. CT findings as a significant predictive factor for the curability of mandibular osteomyelitis: a multivariate analysis. *Dentomaxillofacial Radiol* 2005;34:86–90.
21. Langley-Hobbs SJ, Carmichael S, Lamb CR, Bjornson AP, Day MJ. Polyostotic lymphoma in a young dog: a case report and literature review. *J Small Anim Pract* 1997;38:412–416.
22. Hoenerhoff MJ, Kiupel M, Rosenstein D, Pool RR. Multipotential osteosarcoma with various mesenchymal differentiations in a young dog. *Vet Pathol* 2004;41:264–268.
23. Marec-Berard P, Philip T. Ewing sarcoma: the pediatrician's point of view. *Pediatr Blood Cancer* 2004;42:477–480.
24. Windfuhr JP. Primitive neuroectodermal tumor of the head and neck: incidence, diagnosis and management. *Ann Otol Rhinol Laryngol* 2004;113:533–543.
25. De Cock HEV, Busch MDM, Fry MM, Mehl M, Bollen AW, Higgins RJ. A peripheral primitive neuroectodermal tumor with generalized bone metastases in a puppy. *Vet Pathol* 2004;41:437–441.
26. Daw NC, Mahmoud HH, Meyer WH, et al. Bone sarcomas of the head and neck in children: the St Jude children's research hospital experience. *Cancer* 2000;88:2172–2178.
27. Mascarenhas L, Peteiro A, Ribeiro CA, et al. Skull osteosarcoma: illustrated review. *Acta Neurochir* 2004;146:1235–1239.
28. Hariya Y, Yuasa K, Nakayama E, Kawazu T, Okamura K, Kanda S. Value of computed tomography findings in differentiating between intraosseous malignant tumors and osteomyelitis of the mandible affecting the masticator space. *Oral Surg Oral Med Oral Pathol Radiol Endod* 2003;95:503–509.
29. Cockshutt JR. Bone infection. In: *Bone in clinical orthopedics*, Sumner-Smith G (ed): New York, NY: Thieme, 2002;205–217.
30. Greene CE, Budberg SC. Musculoskeletal infections. In: Greene CE (ed): *Infectious diseases of the dog and cat*, 3rd ed. Philadelphia: Saunders, 2006;823–830.
31. Singh A, Al Khabori M, Hyder MJ. Skull base osteomyelitis: diagnostic and therapeutic challenges in atypical presentation. *Otolaryngol Head Neck Surg* 2005;133:121–125.
32. Slattey WH III, Brackmann DE. Skull base osteomyelitis. Malignant external otitis. *Otolaryngol Clin North Am* 1996;29:795–806.
33. Sreepada GS, Kwartler JA. Skull base osteomyelitis secondary to malignant otitis externa. *Curr Opin Otolaryngol Head Neck Surg* 2003;11:316–323.
34. Klopp LS, Hathcock JT, Sorjonen DC. Magnetic resonance imaging features of brain stem abscessation in two cats. *Vet Radiol Ultrasound* 2000;41:300–307.
35. Lew DP, Waldvogel FA. Osteomyelitis. *Lancet* 2004;364:369–379.
36. Hudson MC, Ramp WK, Nicholton NC, Williams AS, Nousiainen MT. Internalization of *Staphylococcus aureus* by cultured osteoblasts. *Microb Pathog* 1995;19:409–419.
37. Stewart PS. Mechanisms of antibiotic resistance in bacterial biofilms. *Int J Med Microbiol* 2002;292:107–113.